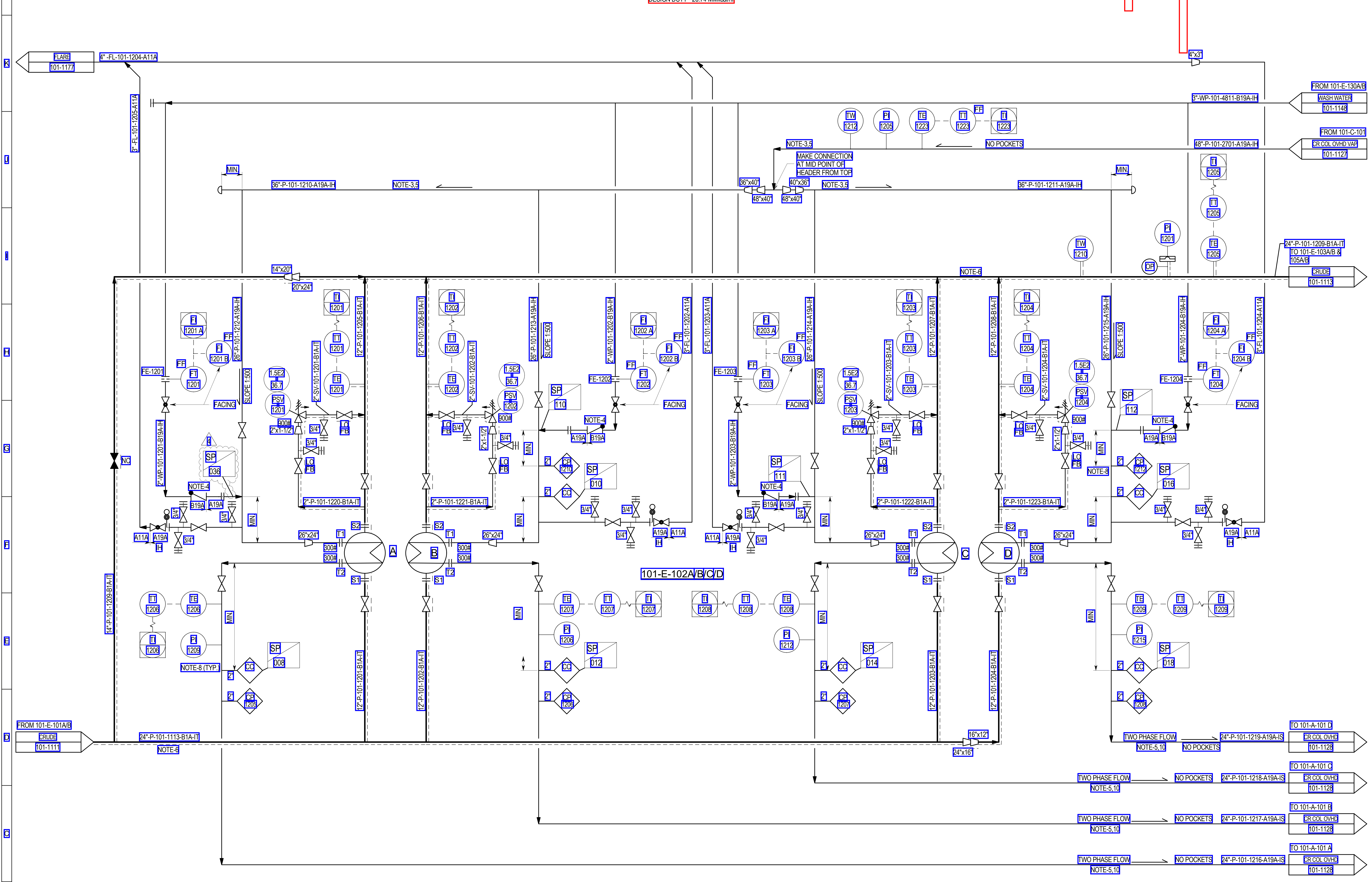
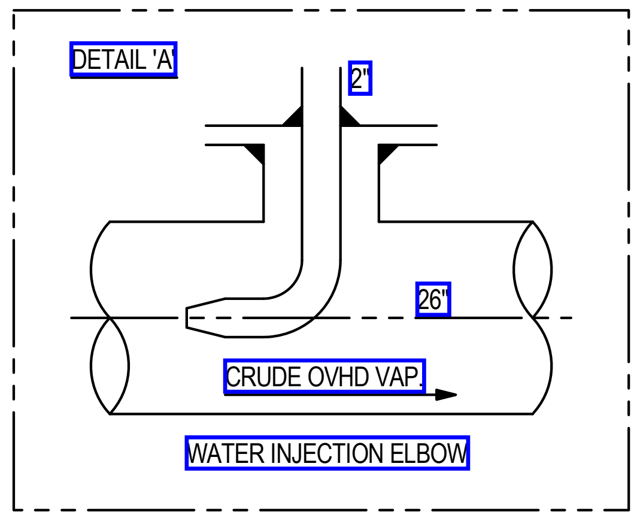
# 101-E-102A/B/C/D
# CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C 4
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr



## NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH `FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED.
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

RED = Equipment Info
BLUE = Process Tags

MW

| संख्या No. | संशोधन REVISIONS | तिथि DATE | ड्रॉन DRN | द्वारा BY | जॉच CKD | अनुमोदित APPD |
|---|---|---|---|---|---|---|
| 4 | ISSUED FOR CONSTRUCTION | 10.03.22 | SC | MW | VD | PV |
| 3 | ISSUED FOR CONSTRUCTION | 30.09.21 | SC | MW | VD | PV |
| 2 | ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION | 02.09.20 | RMB | KG | RS | SG |
| 1 | ISSUED FOR ENGINEERING/DESIGN | 11.08.20 | JO | KG | RS | SG |
| 0 | ISSUED FOR REVIEW | 13.03.20 | LC | KG | RS | SG |

DETAILED ENGINEERING CONTRACTOR: FLUOR®
FLUOR DANIEL INTERNATIONAL
(MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA
TATA PROJECTS LIMITED
MUMBAI

| संख्या No. | तिथि DATE | संशोधन REVISIONS | ड्रॉन DRN | द्वारा BY | जॉच CKD | अनुमोदित APPD |
|---|---|---|---|---|---|---|
| 1 | 14.12.18 | REVISED AS MARKED | PP | UA/AT | GN | AM |
| 0 | 15.11.18 | ISSUED FOR PACKAGE | S.K.L | UA/AT | GN | AM |

PMC :
इंजिनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)
ई.आई.एल
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

OWNER
एचपीसीएल राजस्थान रिफाइनरी लिमिटेड
HPCL Rajasthan Refinery Limited
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पाइपिंग एण्ड इंस्ट्रुमैन्टेशन डायाग्राम
# PIPING AND INSTRUMENTATION DIAGRAM
## CRUDE/VACUUM DISTILLATION UNIT
## CRUDE PREHEAT TRAIN-1B

| अनुमाप SCALE | कार्य संख्या JOB NO. | विभाग DEPT. | अनुभाग SECTN. | इकाई UNIT | आरेख संख्या DWG. No. | संशो REV. |
|---|---|---|---|---|---|---|
| ∽ | B068 | 79 | 41 | 101 | 1112 | 1 |

TPL DRAWING NO. टी पी एल आरेख संख्या: B224-CDU-TPL-101-1112 संशो REV 4

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SIZE A1 (594X841)

DOCUMENT REVIEW DETAILS

☑ 1 No Comments. Doc. B224-CDU-TPL-101-1112 Rev. 4
☐ 2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
☐ 3 Document does not conform to basic requirements as marked. Resubmit for review.
☐ R Retained for Records. ☐ V Void.

Signature with Date 24-AUG-2022
Name of Reviewer UMER AFZAL (C889)
Page No. 1 of 2

Refer file provided against B224-CDU-TPL-101-1111

DOCUMENT REVIEW DETAILS

☑ 1 No Comments. **Doc. B224-CDU-TPL-101-1112 Rev. 4**

☐ 2 Proceed with manufacture / fabrication as per the commented document. Revised document required.

☐ 3 Document does not conform to basic requirements as marked. Resubmit for review.

☐ R Retained for Records. ☐ V Void.

Signature with Date **24-AUG-2022**

Name of Reviewer **UMER AFZAL (C889)**

Page No. **2 of 2**